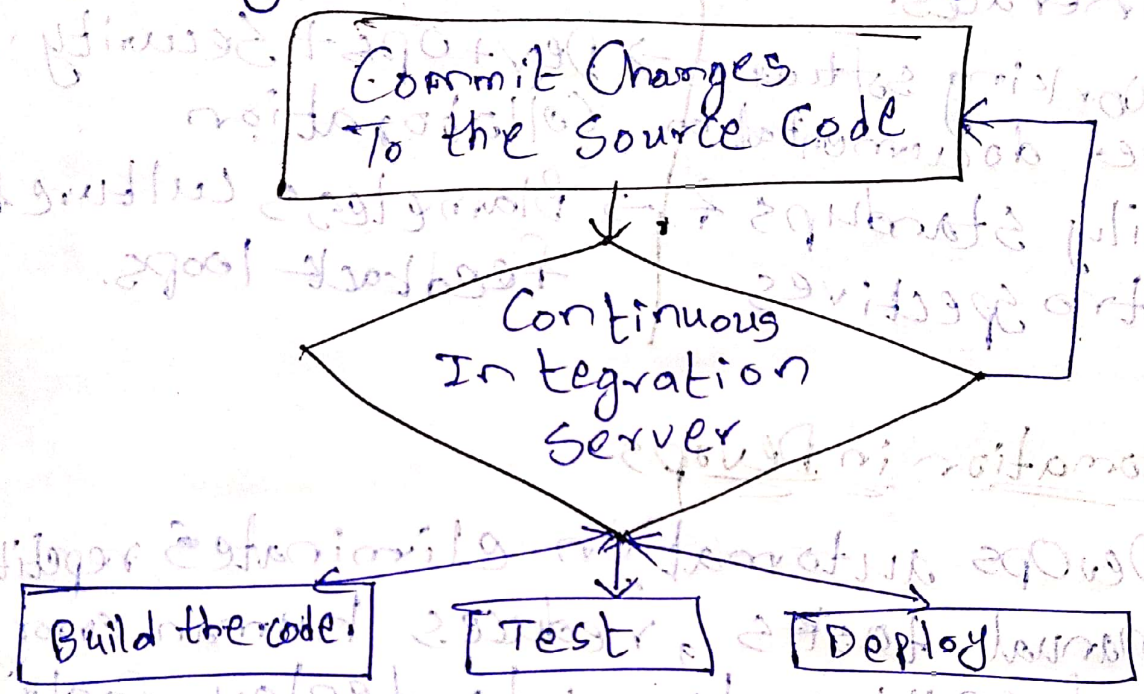
CO-2

Continuous Integration

* The developers are required to commit changes to the source code in a shared repository several times a day or more frequently.



Tools

* Jenkins
* Buildbot
* Travis CI
* Atlassian Bamboo

Companies Implementing DevOps

→ amazon
→ Netflix
→ Target
→ Etsy
→ Walmart

Agile	Devops
→ Iterative development	→ Continuous Integration & Delivery.
→ Customer collaboration over contracts.	→ Automated testing & Deployment
→ Working software over documentation	→ Dev + Ops + Security Collaboration
→ Daily standups & retrospectives	→ Blameless culture & feedback loops.

Automation in Devops

* DevOps automation eliminates repetitive manual tasks, reduces human error, and enables teams to deploy faster & more reliably.
→ CI / CD Automation.
→ Test Automation
→ Security Automation.
→ Container Automation.

ChatOps

* practice of using chat platforms as the central hub for operations.

Infrastructure as Code (IaC)

* This treats servers, networks, & cloud resources exactly like application code.

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Ex:- Terraform, AWS Cloudformation, Ansible.

~~Contain~~ DevOps Best Practices.

→ Shift Left on Security.
→ Use Feature Flags.
→ Practice GitOps.
→ Implement Observability.
→ Blameless Post-Mortems.
→ Small, Frequent Releases.

DevSecOps

*Integrating security into the DevOps pipeline from day one.

GitOps

*This uses Git repositories as the authoritative source for both application code and infrastructure configuration.

CI/CD

*Continuous Integration & Continuous Deployment/Delivery.
*It is a set of rules that automates how software is built, tested & released.
→ Continuous Integration :- Developers merge their code changes into a shared place.

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→ Continuous Delivery / Deployment → Every code change that passes the test goes live automatically.

CI/CD Pipeline

→ Source → Package
→ Build → Release
→ Test → Monitor.
→ Deploy.

CI/CD Tools

→ GitHub Actions
→ Jenkins
→ Docker
→ GitLab CI/CD.

Benefits

→ Faster Software Delivery.
→ Improved Software Quality.
→ Reduced Manual Effort.

Git

* This is a version control system used to track changes in source code during software development.

→ Version Control
→ Distributed System.
→ Branching & Merging.
→ Collaboration.
→ Fast & efficient.

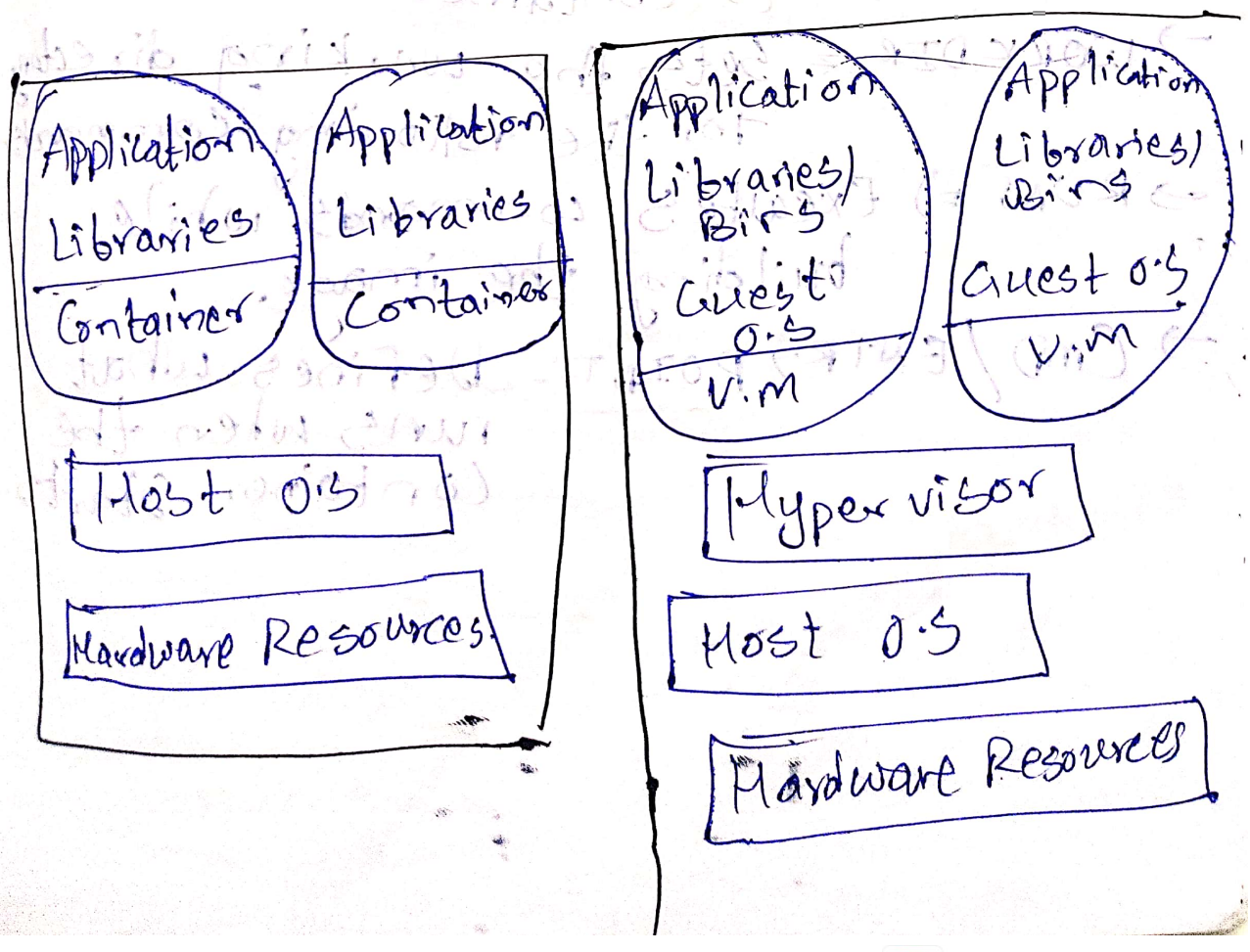
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Containerization Process

* This process begins with application source code & a Dockerfile that specifies how the application should be packaged.
* Then Docker builds an image containing the application & dependencies.
* This image is stored in the registry such as docker hub.

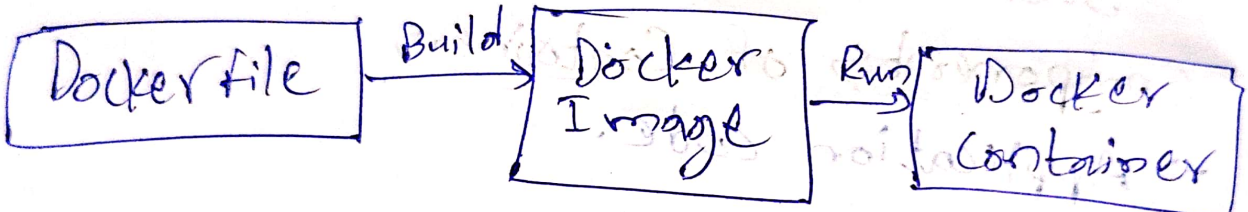
Components of Containers

→ Application code.
→ Libraries.
→ Runtime environments.
→ Namespaces
→ Cgroups ⇒ controls CPU & memory allocation.



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V.M's	Containers
→ Separate Guest O.S	→ Shared Host OS
→ Large (GB's)	→ Small (MB's)
→ Minutes	→ Seconds.
→ Used for multiple O.S	→ Used for microservices.



Dockerfile Instructions

→ FROM = Sets the base image.

→ COPY/ADD = Copies files from host to container.

→ WORKDIR = Sets the working directory for the following commands

→ RUN = Executes commands while building the image.

→ CMD/ENTRYPOINT = Defines what runs when the container starts.

Kubernetes (k8's)

* Manages hundreds of Docker containers manually.
* It automates scheduling, scaling, load balancing, and recovery of containers.
* It ensures applications remain available even when individual containers or servers fail.

Ex:- Netflix, Spotify.

* A pod is the smallest unit in kubernetes.

OWASP (Open Worldwide Web Application Security Project)

* Makes software security visible so that individuals & organizations worldwide can make informed decisions about software risks.

Vulnerable components

→ open-source libraries
→ Frameworks.
→ Plugins
→ API's
→ Containers.

* Old versions remain unpatched.
* Security updates ignored.

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Ex:- Log4 Shell Vulnerability
* Affected millions of Java application.

Prevention
→ Automated vulnerability scanning
→ Timely patch management.

Authentication Failures

Causes:-
→ Weak passwords
→ Missing MFA
→ Credential stuffing.

Session Management Vulnerabilities
* Session ID's exposed in URL's.
* Predictable session tokens.

Software & Data Integrity Failures
→ Due to,
→ Untrusted Plugins
→ Insecure CI/CD pipelines.
→ Unsigned updates

use, Code signing, secure CI/CD pipel
& Integrity monitoring to prevent

Software Supply Chain Attack
Developer
↓
Package Repository
↓
Compromised Package → Application Deployment

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*Malicious code enters production.

Server-Side Request Forgery

* An attacker tricks a server into making unintended requests.

Attacker
↓
Malicious URL
↓
Web Server
↓
Internal Network Resource.

Prometheus

* It is an open-source systems monitoring & altering toolkit designed for reliability & scalability.

→ Time-series Database
→ Multi-Dimensional Data Model.
→ Powerful Query Language (PromQL)

Metrics collection in Prometheus

→ Counter
→ Gauge
→ Histogram
→ Summary.

Grafana

*This is a open-source data visualization & analytics web application.
*Transforms raw data into customizable, real-time dashboards.

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